HW 0526

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1.

(a)

$$\begin{aligned} P(\text{AUTO} = 1 \mid \text{DTIME} = 1) &= \Lambda(-0.2376 + 0.5311(1)) \\ &= \Lambda(0.2935) = \frac{1}{1 + e^{-0.2935}} = \frac{1}{1 + 0.7456} \approx 0.5729 \end{aligned}$$

(b)

$$P(\text{AUTO} = 1 \mid \text{DTIME} = 1) = \Phi(-0.0644 + 0.3000(1)) = \Phi(0.2356) = 0.5931$$

The estimated probabilities from the logit model (0.5729) and the probit model (0.5931) are very close to one another. This is typical for binary choice models because the logistic and standard normal distributions have very similar shapes near the center of the distribution.

(c)

$$\frac{dp}{d\text{DTIME}} = \lambda(\tilde{\gamma}_1 + \tilde{\gamma}_2\text{DTIME}) \cdot \tilde{\gamma}_2, \lambda(z) = \Lambda(z)(1 - \Lambda(z))$$

$$\Lambda(-0.2376 + 0.5311(3)) = \Lambda(1.3557) = \frac{1}{1 + e^{-1.3557}} = \frac{1}{1 + 0.2578} \approx 0.7951$$

$$\lambda(1.3557) = 0.7951 \times (1 - 0.7951) = 0.7951 \times 0.2049 \approx 0.1629$$

$$\frac{dp}{d\text{DTIME}} = 0.1629 \times 0.5311 \approx 0.0865$$

(d)

Logit:

$$
\begin{aligned}
z &= \tilde{\gamma}_1 + \tilde{\gamma}_2\text{DTIME} \\
&= -0.2376 + 0.5311(-5) \\
&= -2.8931
\end{aligned}
$$

$$
\begin{aligned}
\Lambda(-2.8931) &= \frac{1}{1+e^{-(-2.8931)}} = \frac{1}{1+18.0490} \approx 0.0525 \\
\lambda(-2.8931) &= \Lambda(-2.8931)\big(1-\Lambda(-2.8931)\big) \\
&= 0.0525 \times (1-0.0525) \\
&= 0.0525 \times 0.9475 \approx 0.0497
\end{aligned}
$$

$$
\begin{aligned}
\frac{dp}{d\text{DTIME}} &= \lambda(-2.8931)\cdot\tilde{\gamma}_2 \\
&= 0.0497 \times 0.5311 \\
&\approx 0.0264
\end{aligned}
$$

Probit:

$$
\begin{aligned}
z &= \tilde{\beta}_1 + \tilde{\beta}_2\text{DTIME} \\
&= -0.0644 + 0.3000(-5) \\
&= -1.5644
\end{aligned}
$$

$$
\begin{aligned}
\phi(-1.5644) &= \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(-1.5644)^2} \\
&= 0.3989 \times e^{-1.2237} \\
&= 0.3989 \times 0.2941 \\
&\approx 0.1173
\end{aligned}
$$

$$
\begin{aligned}
\frac{dp}{d\text{DTIME}} &= \phi(-1.5644)\cdot\tilde{\beta}_2 \\
&= 0.1173 \times 0.3000 \\
&\approx 0.0352
\end{aligned}
$$

The logit estimate (0.0264) and the probit estimate (0.0352) are both quite small. This happens because at DTIME $= -5$, the individual has a very low baseline probability of driving. However, the probit marginal effect is slightly larger here because the normal distribution density tails off faster than the logistic distribution tails. When evaluating out in the tails (far away from $z = 0$), logit and probit estimates often show slight visible differences. The logit estimate (0.0264) and the probit estimate (0.0352) are both quite small. This happens because at DTIME $= -5$, the individual has a very low baseline probability of driving. However, the probit marginal effect is slightly larger here because the normal distribution density tails off faster than the logistic distribution tails. When evaluating out in the tails (far away from $z = 0$), logit and probit estimates often show slight visible differences.

4.

(a)

$$z = \tilde{\gamma}_1 + \tilde{\gamma}_2 x = -1.836 + 3.021(1.5) = -1.836 + 4.5315 = 2.6955$$
$$P(y = 1 \mid x = 1.5) = \Lambda(2.6955) = \frac{1}{1 + e^{-2.6955}} \approx \frac{1}{1 + 0.0675} \approx 0.9368$$

(b)

Since $0.9368 \geq 0.5$. Predict $\hat{y} = 1$, match $y = 1$.

(c)

$$(x_1, y_1) = (1.5, 1), (x_2, y_2) = (0.6, 1), (x_3, y_3) = (0.7, 0)$$
$$L(\gamma_1, \gamma_2) = \Lambda(z_1) \cdot \Lambda(z_2) \cdot [1 - \Lambda(z_3)]$$

$$\begin{aligned} L(-1, 2) &= \Lambda(-1 + 2(1.5)) \cdot \Lambda(-1 + 2(0.6)) \cdot [1 - \Lambda(-1 + 2(0.7))] \\ &= \Lambda(2) \cdot \Lambda(0.2) \cdot [1 - \Lambda(0.4)] \\ &= 0.8808 \times 0.5498 \times (1 - 0.5987) \\ &= 0.8808 \times 0.5498 \times 0.4013 \approx \mathbf{0.1943} \end{aligned}$$

Real MLE:

$$\ln(L) = -1.612 \implies L = e^{-1.612} = 0.1995$$

The probability function value (0.1995) under the MLE (most likely estimate) is larger than the probability function value (0.1943) for $\gamma_1 = -1, \gamma_2 = 2$.

(d)

True. The likelihood function L for the logit model is a joint probability, which is calculated as the product of the probabilities of observing each individual outcome in the sample. For any binary response model, the logit cumulative distribution function $\Lambda(\cdot)$ yields values strictly between 0 and 1 (i.e., $0 < \Lambda(\cdot) < 1$) for any finite parameter values. Since the total likelihood function is a multiplication of these individual probabilities (or their complements, $1 - \Lambda(\cdot)$), multiplying multiple numbers that are strictly between 0 and 1 will always yield a product that is also between 0 and 1. Therefore, the value of the likelihood function will always satisfy $0 < L < 1$.

(e)

True. The log-likelihood function ($\ln L$) is obtained by taking the natural logarithm of the likelihood function (L). As established in part (d), the value of the likelihood function L represents a joint probability and strictly lies between 0 and 1 ($0 < L < 1$) under any normal estimation. Mathematically, the natural logarithm of any number strictly between 0 and 1 is always negative ($\ln(L) < 0$ for $0 < L < 1$). Therefore, the value of the log-likelihood function will always be negative.

18.

(a)

Table 1: (a) lpm model

	Dependent variable:
lvr	0.002^{**}
	(0.001)
ref	-0.059^{**}
	(0.024)
insur	-0.482^{***}
	(0.030)
rate	0.034^{***}
	(0.010)
amount	0.024
	(0.014)
credit	-0.0004^{**}
	(0.0002)
term	-0.013^{***}
	(0.004)
arm	0.128^{***}
	(0.028)
Constant	0.688^{***}
	(0.229)
Note:	*p<0.1; **p<0.05; ***p<0.01

The signs of the estimated coefficients in the linear probability model are highly reasonable and align perfectly with economic and financial intuition regarding mortgage risk. Specifically, factors that increase a borrower's financial burden or leverage—such as a higher loan-to-value ratio (lvr), higher initial interest rates (rate), and adjustable-rate mortgages (arm), which were notorious for payment shocks during the 2008 subprime crisis—all significantly increase the probability of delinquency. Conversely, factors that enhance a borrower's financial capacity or reduce monthly payment stress—including higher credit scores (credit), refinancing to secure better terms (ref), carrying mortgage insurance to mitigate lender risk (insur), and longer loan maturities (term) that spread out principal payments—all exhibit negative coefficients, indicating a significant decrease in the likelihood of missing payments.

(b)

Yes, the signs of the estimated coefficients are identical across both models, and their statistical significance remains highly consistent with only minor shifts in significance thresholds. Specifically, all variables maintain their directional impact: risk-increasing factors (lvr, rate, amount, arm) retain positive signs, while risk-mitigating factors (ref, insur, credit, term) retain negative signs across both frameworks. In terms of significance, key structural determinants like mortgage insurance (insur), interest rate (rate), loan term (term), and adjustable-rate status (arm) remain highly significant ($p < 0.01$) in both models, and refinancing (ref) remains stable at the 5% level ($p < 0.05$); the only slight differences are that credit score (credit) and loan-to-value ratio (lvr) drop marginally from the 5% to the 10% significance level when moving from the LPM to the Logit model, while loan amount (amount) shifts conversely from being insignificant to marginally significant at the 10% level.

(c)

For both the Linear Probability Model (LPM) and the Logit model, the predicted values represent the estimated probability that a borrower will face mortgage delinquency based on their individual profile. For the 500th observation, the LPM predicts a 18.28% chance of delinquency, while the Logit model predicts a slightly lower 13.36% chance; both frameworks classify this borrower as a low-to-moderate risk who is highly likely to keep up with payments. In contrast, for the 1000th observation, the LPM predicts a 57.85% chance of delinquency, while the Logit model predicts a higher 62.66% chance; because both values cross the 50% threshold, both models flag this applicant as a high-risk individual more likely to default than to repay. The discrepancy between the models stems from their functional forms: the LPM assumes a strictly linear relationship where risk increases at a constant rate, whereas the Logit model utilizes an S-shaped logistic curve that captures non-linear risk acceleration and structurally guarantees that all predicted probabilities stay bounded strictly between 0 and 1.

(d)

Table 2: (b) logit model

	Dependent variable:
	delinquent
lvr	0.014*
	(0.008)
ref	−0.525**
	(0.230)
insur	−3.122***
	(0.217)
rate	0.309***
	(0.083)
amount	0.218*
	(0.114)
credit	−0.004*
	(0.002)
term	−0.133***
	(0.035)
arm	1.406***
	(0.371)
Constant	1.665
	(1.992)
Observations	1,000
Log Likelihood	−333.146
Akaike Inf. Crit.	684.292
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 3: Comparison of Predicted Probabilities of Delinquency

Observation	Linear Probability Model (LPM)	Logit Model
Obs 500	0.1828	0.1336
Obs 1000	0.5785	0.6266

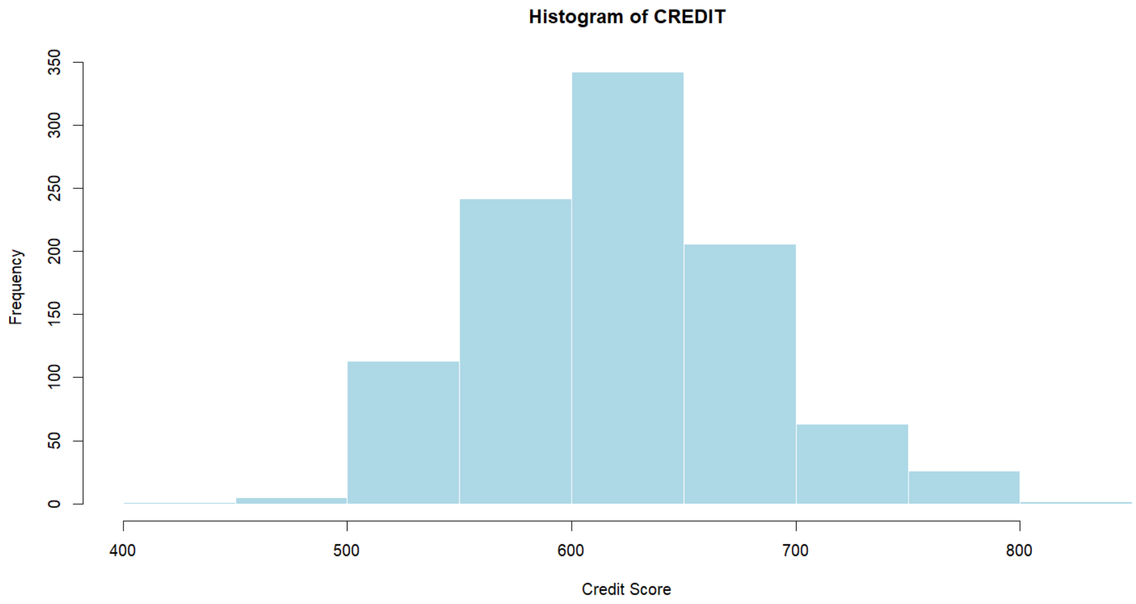

Table 4: Predicted Probabilities of Delinquency by Credit Score

Credit Score (`CREDIT`)	Linear Probability Model (LPM)	Logit Model
500	0.5533	0.4967
600	0.5091	0.4086
700	0.4649	0.3261

Similarities: Both models exhibit a clear negative relationship between credit scores and mortgage delinquency. As CREDIT increases from 500 to 700, the predicted probability of default steadily declines in both frameworks, aligning with the economic intuition that higher creditworthiness implies lower default risk. Furthermore, at the lower credit tier (CREDIT = 500), both models place the borrower right around the critical 50% risk threshold (55.33% for the LPM vs. 49.67% for Logit), consistently flagging low credit scores as highly precarious.
Differences: The fundamental difference lies in their functional forms, which dictate how the probabilities change. The LPM assumes a strictly linear relationship, resulting in a constant additive decrease of exactly 4.42% for every 100-point increase in credit score across all tiers ($\Delta P = 0.5091 - 0.5533 = -0.0442$). Conversely, the Logit model captures a non-linear relationship via an S-shaped curve, meaning the marginal risk reduction varies depending on the baseline score. In this case, moving from 500 to 600 drops the risk by 8.81% (0.4086 − 0.4967), while moving from 600 to 700 drops it by 8.26% (0.3261 − 0.4086). Consequently, within this segment of the data, the Logit model is far more sensitive to credit improvements, providing a much lower and more realistic default probability (32.61%) for prime borrowers (CREDIT = 700) compared to the rigid and overly pessimistic estimate of the LPM (46.49

(e)

The marginal effect measures the change in the probability of mortgage delinquency resulting from a one-unit increase in the borrower's credit score, holding all other explanatory variables constant. In the Linear Probability Model, the marginal effect is inherently

Table 5: Marginal Effects of Credit Score on Delinquency Probability

Credit Score (`CREDIT`)	Linear Probability Model (LPM)	Logit Model
500	−0.000442	−0.000891
600	−0.000442	−0.000861
700	−0.000442	−0.000783

constant across all credit scores, remaining at −0.000442; this means that regardless of whether an applicant has excellent or poor credit, gaining one additional credit point always reduces their delinquency probability by exactly 0.0442%. In contrast, the Logit model's marginal effect varies non-linearly depending on the baseline risk, calculated as $\beta_{\text{credit}} \times \hat{p}(1-\hat{p})$. The Logit results reveal that a one-point increase in credit score reduces default probability by 0.0891% at a score of 500, 0.0861% at 600, and drops to 0.0783% at 700. This reflects the S-shaped logistic distribution, where marginal effects peak near the 50% risk threshold (where uncertainty is highest around CREDIT = 500) and naturally taper off as the borrower's profile moves toward the safer tail of the distribution (CREDIT = 700), providing a more realistic depiction of diminishing returns on credit risk mitigation.

(f)

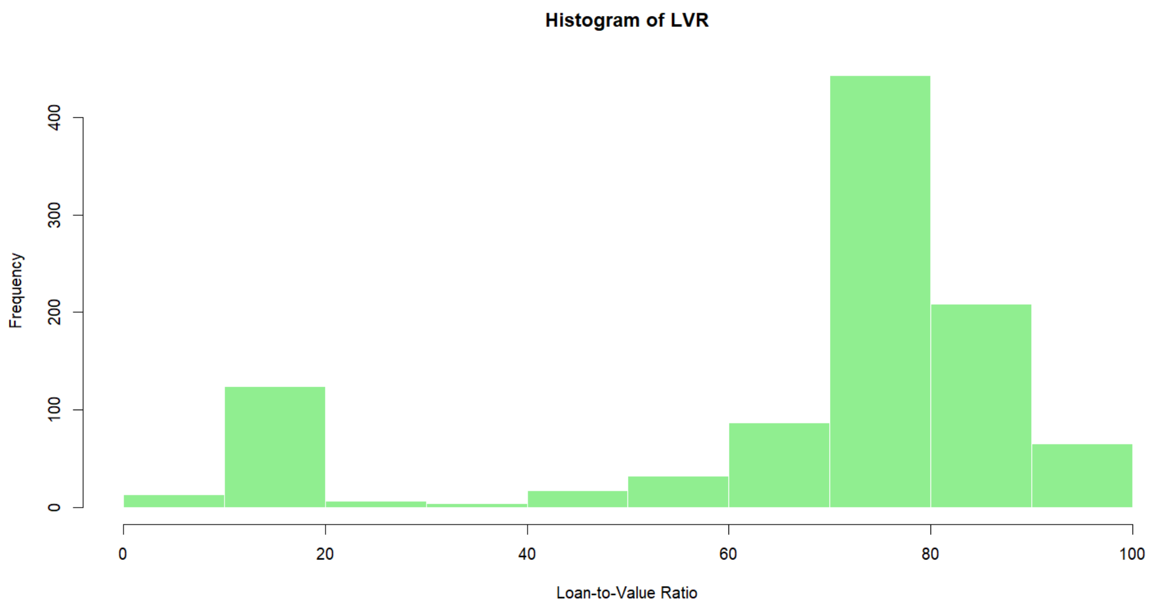

Table 6: Predicted Probabilities of Delinquency by Loan-to-Value (LVR) Ratio

Loan-to-Value Ratio (`LVR`)	Linear Probability Model (LPM)	Logit Model
20%	0.4117	0.2349
80%	0.5091	0.4086

Directional Agreement: Both models strongly agree that higher leverage increases mortgage risk. As LVR jumps from 20% to 80%, the predicted probability of delinquency

rises significantly in both frameworks. This aligns with standard credit risk theory: borrowers with minimal equity (LVR = 80%) are far more vulnerable to defaulting if property values decline.
Discrepancy in Risk Magnitude: The starkest contrast occurs at the safer equity level (LVR = 20%). The LPM predicts a relatively high delinquency probability of 41.17%, whereas the Logit model predicts a much lower and more realistic 23.49%. Because the LPM forces a straight line across all data points, it heavily overestimates the default risk for safe, low-leverage applications.
Linear vs. Non-linear Acceleration: The LPM exhibits a rigid, constant marginal effect where a 60-percentage-point increase in LVR increases the default probability by exactly 9.74% ($\Delta P = 0.5091 - 0.4117 = 0.0974$). In contrast, the Logit model features an S-shaped curvature that captures an accelerated risk surge of 17.37% ($\Delta P = 0.4086 - 0.2349 = 0.1737$) over the same interval. This demonstrates that the Logit model is far more sensitive to structural shifts in borrower leverage, correctly recognizing that jumping to an 80% loan-to-value ratio exposes the lender to a disproportionately steeper escalation in default probability than a linear model can capture.

(g)

Table 7: Comparison of Classification Accuracy (Threshold = 0.5)

Model	Number of Observations	Percentage of Correct Predictions
Linear Probability Model (LPM)	1,000	85.8%
Logit Model	1,000	85.4%

Using a standard classification threshold of 0.5, both models exhibit a highly similar and remarkably high overall accuracy rate, with the Linear Probability Model (LPM) correctly predicting 85.8% of the outcomes and the Logit model correctly predicting 85.4%.While this minor 0.4% difference might superficially suggest that the LPM performs slightly better, evaluating a binary choice model solely on its aggregate "percentage of correct predictions" can be highly misleading due to sample imbalance. In mortgage datasets from this era, the vast majority of borrowers do not default ($Y = 0$).

(h)

No, a 0.5 threshold is not the best rule because it assumes the cost of making a mistake is equal in both directions. In real-world banking, the costs are heavily asymmetric:Type I Error (False Alarm): Rejecting a good borrower only costs you the missed interest profit.Type II Error (Missed Default): Approving a borrower who defaults can cost you the entire loan principal, which is drastically more expensive.Because a 0.5 threshold is too loose, it will mistakenly approve high-risk applicants who have a 35% to 49% chance of defaulting, exposing the bank to severe losses.A Better RuleA superior approach is to lower the decision threshold to an Expected Loss Minimization Threshold (p^*), calculated as:

$$p^* = \frac{\text{Cost of Type I Error}}{\text{Cost of Type I Error} + \text{Cost of Type II Error}}$$

Since the cost of default in the denominator is massive, the optimal threshold p^* drops significantly—typically between 0.10 and 0.20 (10% to 20%).Actionable Strategy: Use the first 500 observations to find the exact probability threshold that maximizes net portfolio profit, and then apply that safer, lower threshold (e.g., 15%) to evaluate applications 501—1,000.